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Price dynamics for foods of varied nutritional characteristics

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Security and Nutrition in the World 2025*

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Security and Nutrition in the World 2025***

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Abstract

This paper analyses global and national trends in retail food prices to understand their relationship with food group, processing level, and nutritional quality. For this purpose, we use standardized global data for 2011, 2017 and 2021, as well as national price series from Mexico, Nigeria and Pakistan between 2019 and 2024. The global analysis suggests three stable patterns. First, basic starchy staples are typically the cheapest source of calories around the world. In contrast, animal-source foods, fruits and vegetables have higher (though more variable) prices. Second, processing affects prices in different ways. In general, ultra-processed foods have on average a lower cost per calorie. However, this is likely because they tend to be in low-cost food groups (starchy staples, sweets, or oils and fats), while unprocessed or minimally processed foods are more likely to be in higher-cost food groups (animal-source foods, fruits and vegetables). Third, nutritional composition is not consistently related to prices in most food groups, though this relationship varies for animal-source foods, fruits, and vegetables. The country-level analysis suggests that food inflation was higher than overall inflation between 2019 and 2024. In all three countries, prices for basic starchy staples rose faster than overall food prices, with particularly harsh implications for low-income households (for whom starchy staples are the most important source of calories to meet their dietary needs). Prices of oils and fats also rose sharply. We also use the national datasets and adopt a modified cost-of-a-healthy-diet approach to assess recent trends in the price of healthy baskets. We find a variety of price trends with declines in Nigeria, seasonal fluctuations in Pakistan, and increases in Mexico. This reveals how the real cost of food and a least-cost healthy diet can vary widely across countries even during periods of high inflation.

Keywords: food prices, inflation, cost of a healthy diet, food processing, nutritional scores.

JEL codes: E31, I31, I12, O15, Q11.

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1. Introduction

Healthy diets, defined as those that align with food-based dietary guidelines, are unaffordable for a large share of the world's population. Since July 2022, FAO and the World Bank have monitored the cost and affordability of healthy diets (CoAHD) using Healthy Diet Basket (HDB) targets (World Bank, 2024). Multiple national governments, including Ethiopia, Ghana, Nigeria and Pakistan, also use the CoAHD metrics to monitor access to healthy diets over time at the national and subnational levels (NBS, 2024; Alemayehu *et al.*, 2023; Herforth *et al.*, 2024). Global and national monitoring efforts have revealed that the cost to purchase nutrient-dense food groups such as fruits, vegetables and animal-source foods is frequently prohibitive for people with low incomes, and in many cases the cost to purchase a healthy diet exceeds the average household's total budget available for food. Global estimates for the cost of a healthy diet reveal that 32 percent of the world's population could not afford to purchase the foods required for health in 2024 (FAO, IFAD, UNICEF, WFP and WHO, 2025).

However, there is abundant evidence that people do not always consume healthy diets even when their available income suggests they could afford to do so (Imamura *et al.*, 2015; Miller *et al.*, 2022). Food supply data indicate that global dietary shortfalls are concentrated in fruits, vegetables, and legumes, nuts and seeds, while diets in South Asia and Sub-Saharan Africa also include inadequate quantities of animal-source foods (Costlow *et al.*, 2025). There are several competing narratives that might explain the observation that people largely do not consume healthy diets. One possibility is that relative prices are higher for healthier items than for less healthy items, either within each food group or across all foods available in a marketplace (Headey and Alderman, 2019), so consumers choose less expensive but less healthy foods. Another possibility is that relative prices are not systematically higher for healthier or unhealthier items, but consumers decisions are swayed by substantial marketing and advertising for less healthy foods (Kirk *et al.*, 2010). A third narrative presents unhealthy foods as irresistible or even addictive to the average consumer (Gearhardt and Schulte, 2021). From this perspective, people are inherently vulnerable to the influence of highly palatable foods that are high in added sugar, sodium, or saturated fats. A fourth possibility is that costs of food acquisition, storage, preparation and consumption are higher for healthier diets, prompting consumers to choose foods whose post-retail costs are more affordable (IOM and NRC, 2013).

Recent high rates of food price inflation provide additional motivation to address the question of whether food prices are systematically higher for healthier foods (FAO, 2025a; FAO, 2025b). Inflation represents a loss of purchasing power for people whose incomes do not grow at a rate that matches price increases, and has been shown to cause worse outcomes for nutrition and health, especially among young children and other vulnerable populations (Headey and Ruel, 2023; Kidane and Woldemichael, 2020; Lee *et al.*, 2016; Movsisyan *et al.*, 2024). Evidence from recessions following the 2008 financial crisis and the COVID-19 pandemic show that poor consumers typically experience higher inflation rates than published inflation statistics due either to their constrained ability to shift spending or higher inflation rates among low-cost goods (Argente and Lee, 2021; Cavallo and Krytsov, 2024). These patterns increase the risk of welfare reduction that low-income consumers face during inflationary periods.

Food inflation may have further differential implications for nutrition and welfare if not equally distributed across all foods. Food inflation concentrated in more affordable foods or in healthier foods at all price points may harm nutrition and diet-related health outcomes by reducing

access to nutritious and healthy diets. In contrast, food inflation concentrated in foods that are more expensive or less healthy will have less impact on access to healthy diets. The degree to which a food item is processed may also be associated with food price inflation, as food manufacturing requires the addition of labour and inputs during the food production process (FAO *et al.*, 2011). The added cost of labour and inputs may be passed through to the retail consumer to some degree, resulting in higher prices than would be observed for the unprocessed commodity ingredients. Alternatively, processed foods could be less expensive or exhibit lower price volatility due to longer shelf life, which may reduce rates of spoilage while allowing for arbitrage in wholesale markets and storage by the end user. At any given time, food prices may also vary systematically across geographic regions and national income levels. In countries where food manufacturing capacity is not highly developed, prices for processed and ultra-processed foods within the same food groups may exceed those for unprocessed foods by a larger margin than in countries where food manufacturing has achieved economies of scale.

This background paper presents a systematic analysis of retail food price data at the global level and in three selected country case studies. The objective of the two analyses is to determine whether food prices and price premiums for recommended food groups vary systematically across a range of nutritional attributes of food, and whether these price patterns have changed over time. We use global retail food prices from the International Comparison Program (ICP), coordinated by the World Bank, and national prices from Mexico, Nigeria and Pakistan. We investigate cross-sectional patterns in global food prices and trends over time in national data to assess whether dispersion in food prices is predicted by key nutritional characteristics of food, and whether the underlying patterns in price dispersion across nutritional characteristics were stable throughout the post-pandemic inflationary period. To do so, we compare prices by food group, level of processing, and healthfulness as measured by Food Compass Score with average prices for basic starchy staples. These analyses provide multiple perspectives on the central question of whether healthier foods cost more than less healthy foods as well as its corollary: whether high prices are the primary barrier to consumption of healthy diets.

2. Background

2.1 Variation in nutritional quality of retail food items

2.1.1 Food prices and nutritional quality

Food prices differ significantly between each of the food groups recommended for consumption in healthy diets. Using global food price monitoring data from ICP, Headey and Alderman (2019) show that starchy staples, cooking oils and sugars are less expensive per unit of dietary energy than nutrient-dense food groups such as fruits, vegetables and animal-source foods (Headey and Alderman, 2019). The comparatively low cost of starchy staples is the basis for Bennett's Law, which describes how lower-income populations have long been observed to obtain a larger share of their dietary energy from that food group. The low cost per unit of energy from cooking oils and sugars is a more recent result of industrialization and globalization that has reduced the costs of production and distribution faster for these foods than for more nutrient-dense food groups that are typically bulkier and more perishable, and also have lower yields in terms of calories per hectare (Masters and Finaret, 2024). Low-income consumers also tend to allocate a large share of food spending to starchy staples, as shown in a sample of 11 African countries for *The State of Food Security and Nutrition in the World 2023* (FAO, IFAD, UNICEF, WFP and WHO, 2023). Recent studies from both Mexico and the United States of America show that fruits, vegetables and meat are significantly more expensive per unit of dietary energy than staple grains and sweet and salty grain products (Batis *et al.*, 2022; Carlson and Frazao, 2012). However, when comparing the price of foods within the same food group, higher prices are not consistently associated with higher nutritional profile (Martinez *et al.*, 2024). In some food groups – such as vegetables, fruits, and legumes, nuts and seeds – nearly all options are highly nutritious, so there are healthful options available at all price points. In contrast, there is more variability in the healthfulness of options available in other food groups, such as animal-source foods. Healthier animal-source foods tend to be more expensive on average, but this relationship is not consistent at all price points, and the most nutritious animal-source foods are not always the most expensive (Martinez *et al.*, 2024).

2.1.2 Diet cost and diet quality

In analyses of diets observed through national dietary intake surveys in Belgium, Mexico, New Zealand and the United States of America, more nutritious diets were more expensive on average (Conrad *et al.*, 2021; Curi-Quinto *et al.*, 2022; Vandevijvere *et al.*, 2018; Vandevijvere, Seck *et al.*, 2020). However, while people who consume healthier diets tend to spend more on food, not all healthier dietary patterns are more expensive. In the same studies of observed diets, some individuals who consumed high quality diets spent comparatively less on food. In Mexico, higher quality diets were more expensive than current diets on average, but healthier diets that were also more environmentally friendly were less expensive than both average diets and other high-quality diets (Curi-Quinto *et al.*, 2022). In New Zealand, healthy diets were more expensive than average. However, the high costs of healthy diets were mainly driven by the expensive food consumed outside the home by many of these individuals. About one quarter of the healthy diets observed were cheaper than average (Vandevijvere *et al.*, 2018). Modelling of theoretical dietary patterns also shows that more nutritious diets are sometimes less expensive. For example, Springmann *et al.* (2021) show that vegetarian and vegan diets, which tend to be more nutritious than average diets, are less expensive than current diets

globally, due to the high share of dietary energy from expensive animal-source foods consumed in middle and higher income countries.

2.2 Variation in processing of retail food items

2.2.1 Trends in consumption of processed foods

Different types of processing methods modify foods for a variety of different purposes. Processing makes foods more appealing by altering their taste and palatability, extends shelf stability through use of preservatives or reduction of water content, and reduces preparation time through partial or complete cooking (Forde and Decker, 2022). Most studies use the NOVA food classification system to categorize foods by level of processing, which distinguishes between unprocessed or minimally processed foods, processed culinary ingredients, processed foods and ultra-processed foods (Monteiro *et al.*, 2018). Consumption of processed foods has increased globally in both urban and rural areas, first in higher-income countries in the late 1900s and later in low- and middle-income countries. Urbanization, rising incomes, and increased work outside of the home drive demand for convenient, appealing processed foods (Nguyen *et al.*, 2021; Popkin and Ng, 2022; Raneri *et al.*, 2019; Reardon *et al.*, 2021; Sauer *et al.*, 2021).

2.2.2 Food prices and processing level

Analyses from dietary intake surveys in Belgium, the Kingdom of the Netherlands, the United Kingdom of Great Britain and Northern Ireland and the United States of America show that ultra-processed foods are both less nutrient dense and less expensive per kilocalorie compared to unprocessed or minimally processed foods (Aceves-Martins *et al.*, 2022; Gupta *et al.*, 2019; Vandevijvere, Pedroni *et al.*, 2020; Vellinga *et al.*, 2022). However, evidence is relatively sparse on the comparative cost of processed foods in low- and middle-income countries. Studies in Vietnam and Mexico show that, as in higher-income countries, ultra-processed foods tend to be much more expensive per kilocalorie than unprocessed staple grains (Batis *et al.*, 2022; Nguyen *et al.*, 2021) but less expensive than some nutrient-dense foods such as fruits, vegetables and meat (Batis *et al.*, 2022). However, in an analysis of dietary intake surveys in Brazil, ultra-processed foods were only slightly more expensive than processed and minimally processed or unprocessed foods (Machado *et al.*, 2017).

2.3 Diet quality, processing level and nutritional outcomes

2.3.1 Diet quality and nutritional outcomes

Poor diets drive a larger proportion of global age-standardized adult mortality and disability-adjusted life years (DALYs) than any other risk factor, driving 22 percent of all deaths and 15 percent of DALYs among adults aged 25 and older (Afshin *et al.*, 2019). The most common diet-related diseases are cardiovascular disease, diabetes and cancers (Afshin *et al.*, 2019). At the same time, inadequate intakes of dietary energy, protein and micronutrients are persistent drivers of morbidity and mortality, particularly for women of reproductive age and young children (Black *et al.*, 2013; FAO, IFAD, UNICEF, WFP and WHO, 2024). Undernutrition causes an estimated 45 percent of child deaths globally, mostly in low- and middle-income countries (Black *et al.*, 2013).

2.3.2 Processing level and nutritional outcomes

The wide spectrum of types of processing means that there is no a priori relationship between NOVA categorization and the healthfulness of a given food. However, many ultra-processed foods have attributes that are associated with poor health. High consumption of ultra-processed foods is associated with a wide variety of poor health outcomes, including higher risk of overweight and obesity, metabolic syndrome, cancer, non-alcoholic fatty liver disease, inflammatory bowel conditions, renal function decline, depression, dementia, diabetes and hypertension (Adjibade *et al.*, 2019; Fiolet *et al.*, 2018; Li *et al.*, 2022; Lo *et al.*, 2022; Moradi *et al.*, 2021; Pagliai *et al.*, 2021; Rey-García *et al.*, 2021; Wang *et al.*, 2022; Zhang *et al.*, 2022). Though this epidemiological evidence strongly suggests that high consumption of ultra-processed food is detrimental for health, there is no consensus on whether these associations are driven by processing level, presence of detrimental food additives, low nutritional quality of ultra-processed foods, or displacement of nutrient-dense foods (Levy *et al.*, 2024; Poti *et al.*, 2017; Reardon *et al.*, 2021).

3. Data and methods

For the global analysis, we use national average retail food prices and availability from ICP in 2011, 2017 and 2021. ICP data are reported to the World Bank by national statistical offices for a standardized list of foods that are commonly available in all countries and in each World Bank geographic region. Food prices are reported in local currencies per reference quantity, as sold at retail. ICP surveys, although generally consistent, may adjust the inclusion of food items across cycles, or other surveying practices or approaches, and are therefore not strictly comparable across years. These differences do not affect the accuracy of purchasing power parity (PPP) estimates, as the PPP calculation does not require a harmonized price matrix over time.

For the country-level analysis, we use food price data collected from national statistical organizations in Mexico, Nigeria and Pakistan. Characteristics of each dataset are described in Table 1. We deflate prices using the general consumer price index (CPI) to ensure that comparisons within and across food groups do not reflect general inflation rates across the economy. All prices are converted to constant July 2019 values, which is the first month of available data. For Nigeria, the state-level CPI, for Pakistan, national CPI at the urban–rural level, and for Mexico national CPI were used. National average prices are converted to 3-month rolling means to reduce the influence of random error.

For both the global and the country-level analysis, we compare average prices within each food category with average prices for basic starchy staples. Basic starchy staples are defined as unprocessed or minimally processed starchy staples, i.e. belonging to NOVA category 1. This includes items such as rice, wheat, flour, noodles and starchy tubers. The ratio of the average price for the food category of interest to the price of basic starchy staples indicates to what degree purchasing different foods imposes a price penalty relative to the cheaper food items relied upon for survival. In both analyses, we assess the magnitude of price dispersion by nutritional category. For the country-level analysis, we plot these price ratios over time to assess whether the underlying price dispersion in each set of comparisons has fundamentally changed during the study period.

For each of the global and country-level analyses, we convert prices per reference unit as sold (e.g. piece or package) to price per kilogram of edible matter, and then to price per kilocalorie, to account for variation in water content within and between food groups, using food item descriptions and food composition tables. For the global analysis, food composition data were retrieved from the United States Department of Agriculture, Standard Reference 28 (USDA SR-28), the FAO/INFOODS Food Composition Table for West Africa, the USDA Food and Nutrient Database for Dietary Studies (FNDDS) 2019–2020, and national food composition tables from Bangladesh, India, Japan and the United Kingdom of Great Britain and Northern Ireland. For the country-level analyses, food composition data were primarily drawn from the USDA SR-28, supplemented by other national databases including the FAO/INFOODS Food Composition Table for West Africa and the Food Composition Table for Pakistan and the USDA Global Branded Foods Database (FAO, 2016; Government of Pakistan Ministry of Planning and Development *et al.*, 2001; USDA, 2023).

Table 1. Food price metadata

	Number of items	Temporal unit	Time period available	Geographic unit
Global analysis				
ICP 2011	893	Year	2011	Country
ICP 2017	852	Year	2017	Country
ICP 2021	735	Year	2021	Country
National analysis				
Mexico	30 030	Month	July 2019–July 2024	Municipality
Nigeria	236	Month	July 2019–July 2024	State, rural
Pakistan	101	Month	July 2019–July 2024	District, rural

Note: ICP = International Comparison Program.

Source: Authors' own elaboration based on World Bank. 2025. International Comparison Program. In: *World Bank*. [Cited 28 May 2025]. <https://www.worldbank.org/en/programs/icp>

To establish categories and benchmark quantities for the foods required for health, we use the Healthy Diet Basket (HDB) approach that has been adopted for global monitoring by FAO and the World Bank. The HDB method was first developed from national food-based dietary guidelines for *The State of Food Security and Nutrition in the World 2020* (Herforth *et al.*, 2020) and subsequently revised for *The State of Food Security and Nutrition in the World 2022* (Herforth *et al.*, 2022). The revised method has since been validated against macro- and micronutrient requirements (Herforth *et al.*, 2024). The HDB framework includes benchmarks for six food groups: starchy staples, oils and fats, fruits, vegetables, animal-source foods, and legumes, nuts and seeds. To facilitate analyses comparing more and less nutritious options within each food group, we include some food items excluded from the cost of a healthy diet (CoHD) estimates that are not recommended in food-based dietary guidelines (Table 2). For example, we include cured meats in the animal-source foods category. We also add a seventh food category comprising sugars, sweets and candies (described in the text as “sweets”), which do not fit in any of the six HDB food categories and are not recommended as part of a healthy diet. We exclude infant foods, mixed dishes, condiments (e.g. vinegar, soy sauce, fish sauce, tomato paste), plants and dairy products typically consumed as condiments (e.g. lemons, limes, garlic, ginger, herbs, sour cream), culinary ingredients (e.g. salt, spices, cornstarch) and alcoholic and non-caloric beverages (e.g. beer, wine, spirits, coffee, tea).

Table 2. Examples of items in cost of a healthy diet by food group

HDB food group	Food items included in CoHD estimates with the Healthy Diet Basket	Food items excluded from CoHD estimates and included in food price analysis
Starchy staples	Rice, wheat, maize, bread, potatoes, plantains, other starchy vegetables.	Potato chips, instant noodles.
Animal-source foods	Meat, fish, milk, dairy, eggs, yogurt, cheese.	Cured meats, chicken nuggets.
Legumes, nuts and seeds	Beans, lentils, nuts, seeds.	
Vegetables	Cabbage, cauliflower, eggplant, pepper, mushroom, spinach, tomato, other green leafy vegetables.	Pickled vegetables, tomato sauce.
Fruits	Apple, banana, strawberry, cherry, mango, orange, pineapple.	Tinned fruits in syrup, fruit juices, sweetened tamarind.
Oils and fats	Maize oil, coconut oil, soybean oil, margarine, other vegetable and seed oils.	Butter, lard.
Sugars, sweets and candies	None.	Cake, muffin, cookie, jam, honey, chocolate, ice cream.
Excluded	None.	Infant foods, condiments, culinary ingredients, alcoholic and non-caloric beverages.

Notes: HDB = Healthy Diet Basket; CoHD = cost of a healthy diet. Items in the left-hand column are recommended as part of a healthy diet and eligible for inclusion in CoHD estimates. Items in the right-hand column are not recommended as part of a healthy diet but are included in analyses of price dynamics by food group.

Source: Authors' own elaboration.

We use the updated Food Compass Score (FCS 2.0) nutritional profile system to quantify the nutritional quality of food items in our analysis (Barrett *et al.* 2024). Food Compass Score rates foods on a scale from 0 to 100 based on nine domains associated with health outcomes: nutrient ratios, vitamins, minerals, food-based ingredients, additives, processing, specific lipids, fiber and protein and phytochemicals (Mozaffarian *et al.* 2021). Foods with lower FCS (lower nutritional profile) have attributes associated with poor health, while foods with higher FCS (higher nutritional profile) have attributes associated with improved health. We use the NOVA Food Classification System to categorize foods by level of processing, and exclude processed culinary ingredients (NOVA 2) for this section of the analysis, as these foods are not typically consumed on their own.

Table 3. NOVA food classification system

NOVA classification	Description	Examples
Unprocessed or minimally processed foods	Edible parts of plants or animals that are unprocessed or processed in ways that preserve freshness, increase shelf stability, or improve food safety .	Fresh, dried or frozen fruits, vegetables, meat, eggs, milk.
Processed culinary ingredients	Ingredients used to make foods more varied and enjoyable that are typically not consumed on their own .	Butter, oil, salt, sugar, spices.
Processed foods	Foods processed by adding culinary ingredients including salt, oils and sugar .	Tinned fruits and vegetables, cheeses, breads.
Ultra-processed foods	Foods made from processed ingredients , typically including ingredients extracted from foods such as preservatives, antioxidants and stabilizers.	Soft drinks, packaged snacks, reconstituted meats.

Source: Monteiro, C.A., Cannon, G., Moubarac, J.-C., Levy, R.B., Louzada, M.L.C. & Jaime, P.C. 2018. The UN Decade of Nutrition, the NOVA food classification and the trouble with ultra-processing. *Public Health Nutrition*, 21(1): 5–17. <https://doi.org/10.1017/S1368980017000234>

4. Results

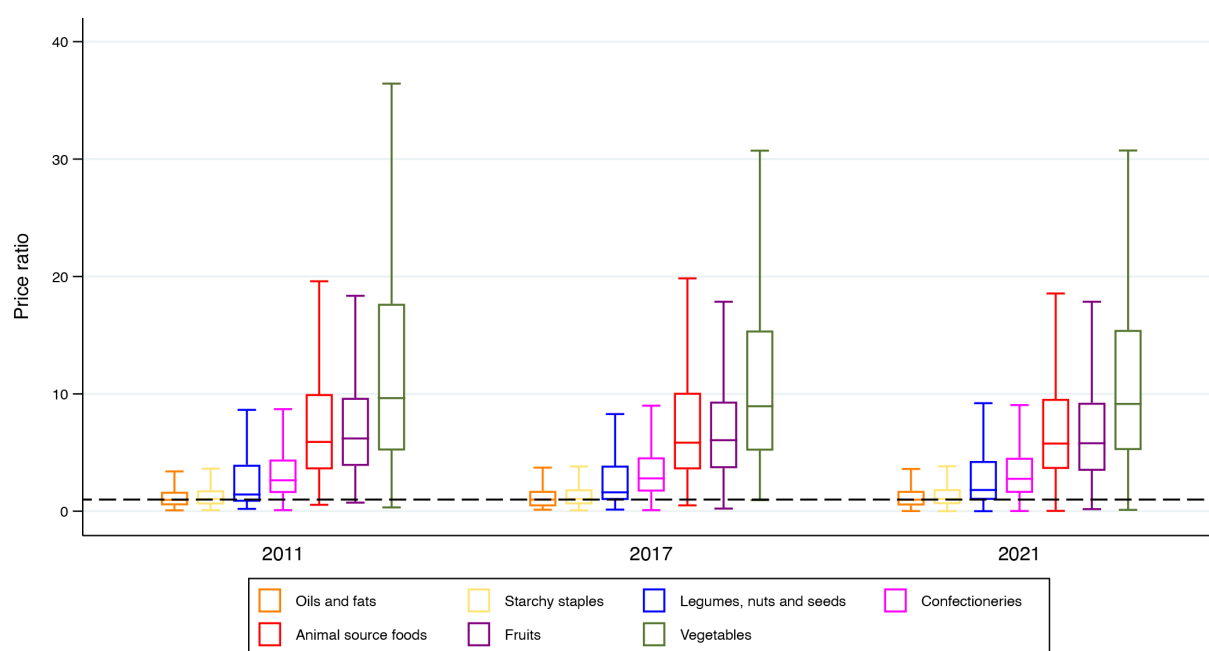
4.1 Global results

When we compare the prices of food items sold at retail around the world, we find that food prices vary by food group, nutritional profile and processing level. These differences are consistent between countries of different national income levels and over the three years of food price data observed (2011, 2017 and 2021).

4.1.1 Price variation by food group

Basic starchy staples (grains, starchy tubers, etc.) are consistently the least expensive form of dietary energy around the world. More nutrient-dense food groups, including vegetables, fruits and animal-source foods, are the most expensive food groups in terms of cost per kilocalorie. On average, vegetables were 11.9 times as expensive as basic starchy staples in 2021, animal-source foods were 9.1 times as expensive, and fruits were 7.2 times as expensive. The difference in price is smaller but still notable for other food groups. Sweets were 3.7 times as expensive as basic starchy staples in 2021, while legumes, nuts and seeds were 3.1 times as expensive. Despite these systematic differences in price by food group, there are items available in all food groups that are much less expensive than basic starchy staples (Figure 1 and Table A4).

Figure 1. Ratio of food prices relative to the price of basic starchy staples, by food group and year

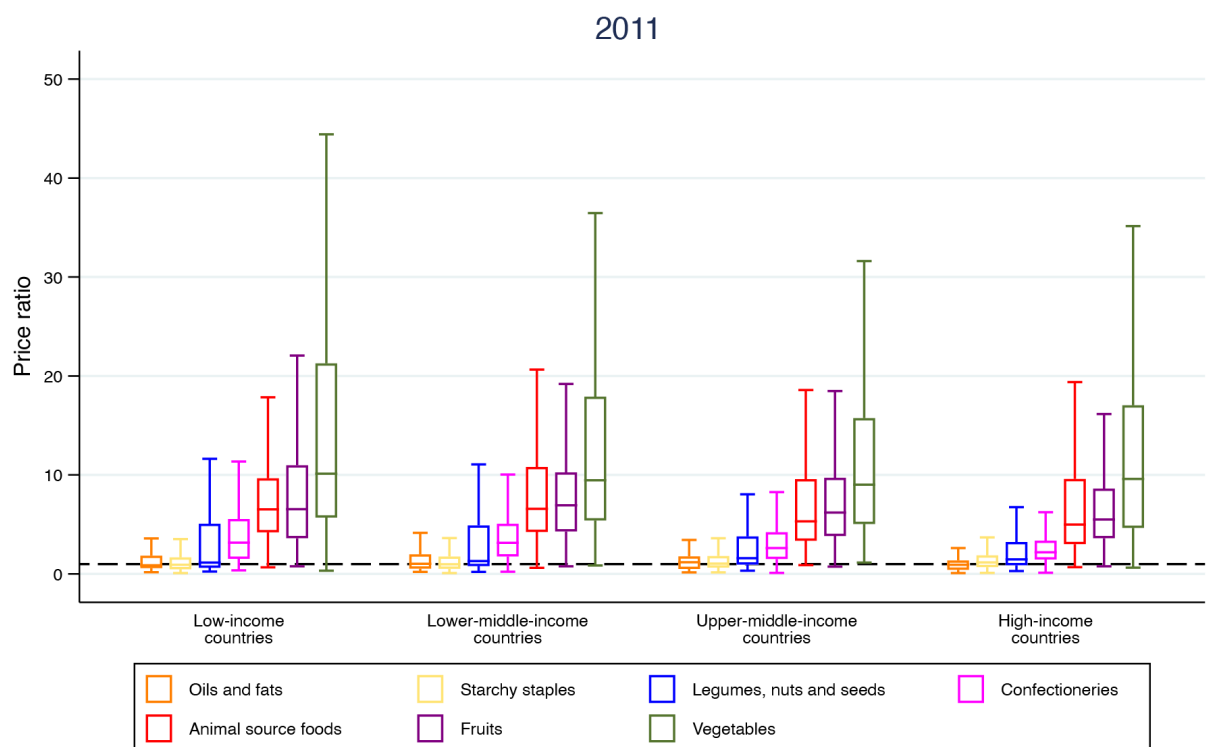


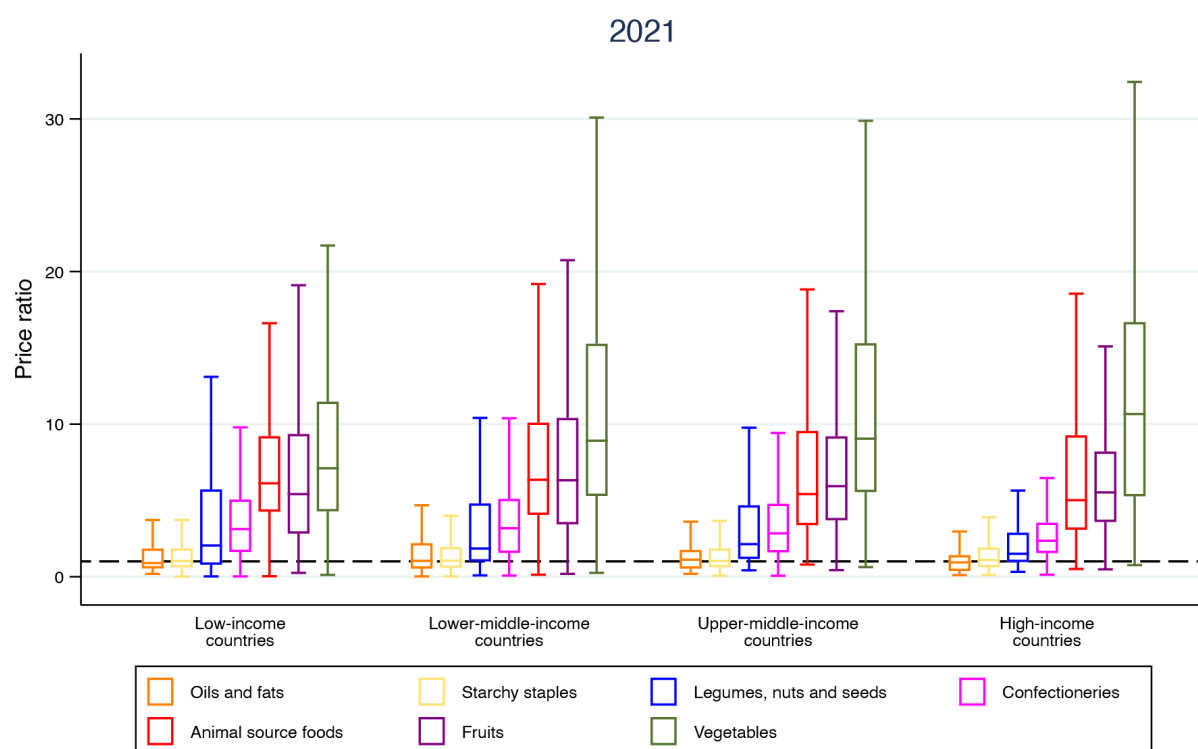
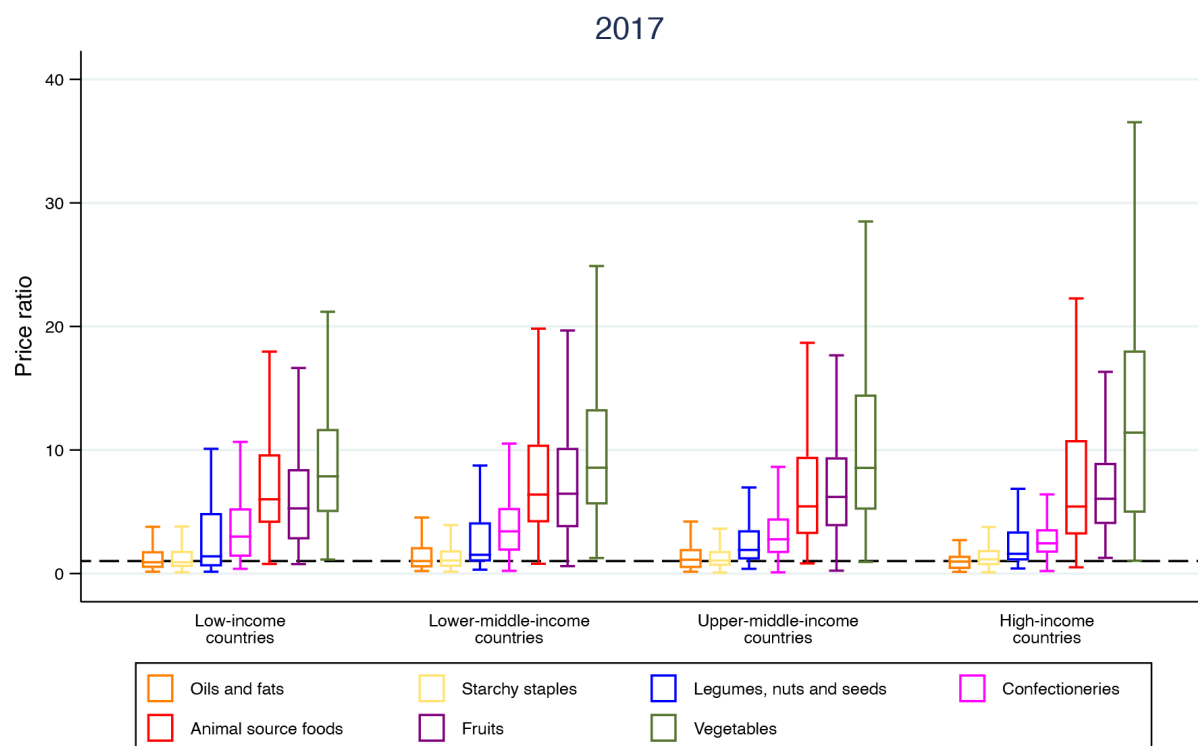
Notes: Data shown are price ratios (price per kilocalorie of edible matter for each food item in each country/year divided by the average price per kilocalorie of basic starchy staples in that country/year). Basic starchy staples are those classified as NOVA 1 (unprocessed or minimally processed foods). Prices are national average retail food prices from ICP in 2011, 2017 and 2021. For each box plot, the centre line indicates the median; the box shows interquartile range (25th to 75th percentile); and the outer lines show the range of data points that fall within 1.5*IQR around the box. Observations that fall outside of 1.5*IQR are not shown. The dashed line shows where the price ratio is equal to 1.

Source: Authors' own elaboration.

These differences in price between food groups are consistent over the three years observed (Figure 1). In addition, this pattern of difference in price between food groups is observed across country income levels. For some food groups, the magnitude of the price ratio varies by country income level. For example, vegetables were less expensive in low-income countries (8.9 times as expensive as basic starchy staples) compared to lower-middle- (12.4 times), upper-middle- (12.3 times) and high-income countries (12.3 times). Sweets were less expensive in high-income countries (3.0 times as expensive as basic starchy staples) compared to upper-middle- (3.9 times), lower-middle- (4.1 times) and low-income countries (4.0 times). These trends were also consistent across the three years of data observed (Figure 2 and Table A4).

Figure 2. Ratio of food prices relative to the price of basic starchy staples, by food group, income level and year





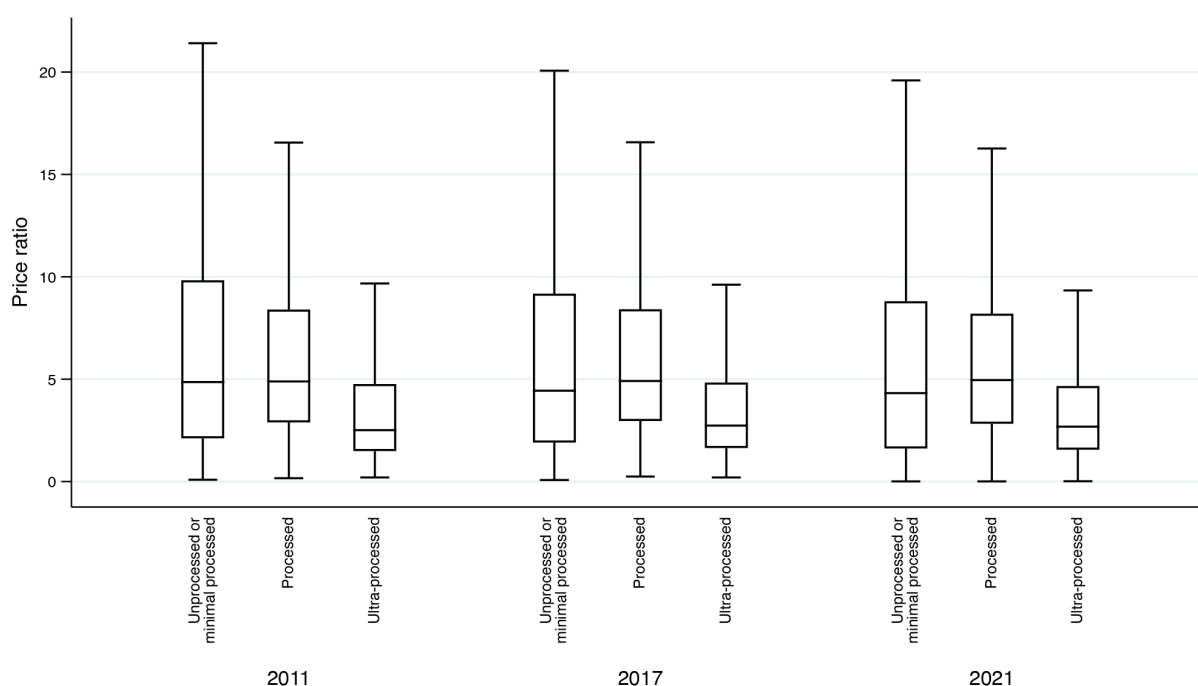
Notes: Data shown are price ratios (price per kilocalorie of edible matter for each food item in each country/year divided by the average price per kilocalorie of basic starchy staples in that country/year). Basic starchy staples are those classified as NOVA 1 (unprocessed or minimally processed foods) or NOVA 3 (processed foods). Prices are national average retail food prices from ICP in 2011, 2017 and 2021. For each box plot, the centre line indicates the median; the box shows interquartile range (25th to 75th percentile); and the outer lines show the range of data points that fall within 1.5*IQR around the box. Observations that fall outside of 1.5*IQR are not shown. The dashed line shows where the price ratio is equal to 1.

Source: Authors' own elaboration.

4.1.2 Price variation by processing level

On average, ultra-processed foods are less expensive per kilocalorie compared to other foods. Considering all retail foods reported by the ICP, ultra-processed foods (NOVA 4) are less expensive per kilocalorie compared to both processed foods (NOVA 3) and unprocessed or minimally processed foods (NOVA 1). In 2021, ultra-processed foods were 3.8 times as expensive as basic starchy staples, while processed foods were 7.5 times as expensive and unprocessed or minimally processed foods were 7.2 times as expensive. This large difference in price per kilocalorie by processing level is observed across all three rounds of data. However, this trend does not hold within food groups. Further, the least expensive options available in many countries are unprocessed (Figure 3 and Table A2).

Figure 3. Ratio of food prices relative to the price of basic starchy staples, by processing level and year



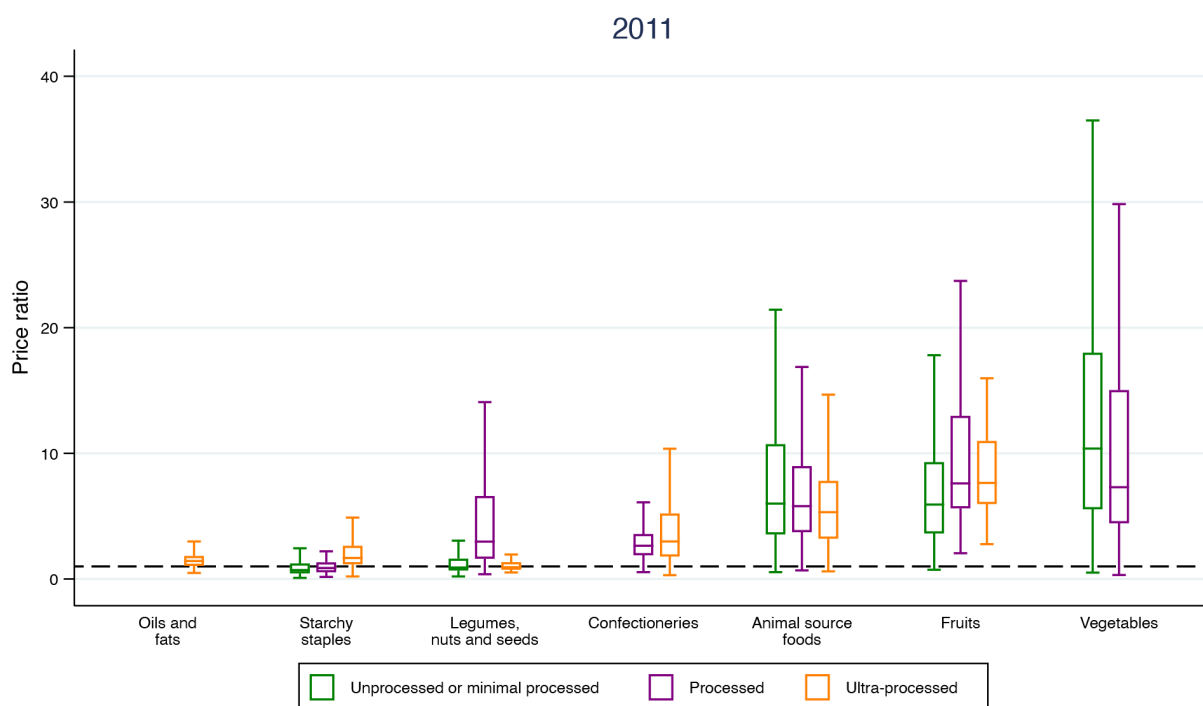
Notes: Data shown are price ratios (price per kilocalorie of edible matter for each food item in each country/year divided by the average price per kilocalorie of basic starchy staples in that country/year). Basic starchy staples are those classified as NOVA 1 (unprocessed or minimally processed foods). Prices are national average retail food prices from ICP in 2011, 2017 and 2021. NOVA categories are shown on the x-axis. For each box plot, the centre line indicates the median; the box shows interquartile range (25th to 75th percentile); and the outer lines show the range of data points that fall within 1.5*IQR around the box. Observations that fall outside of 1.5*IQR are not shown.

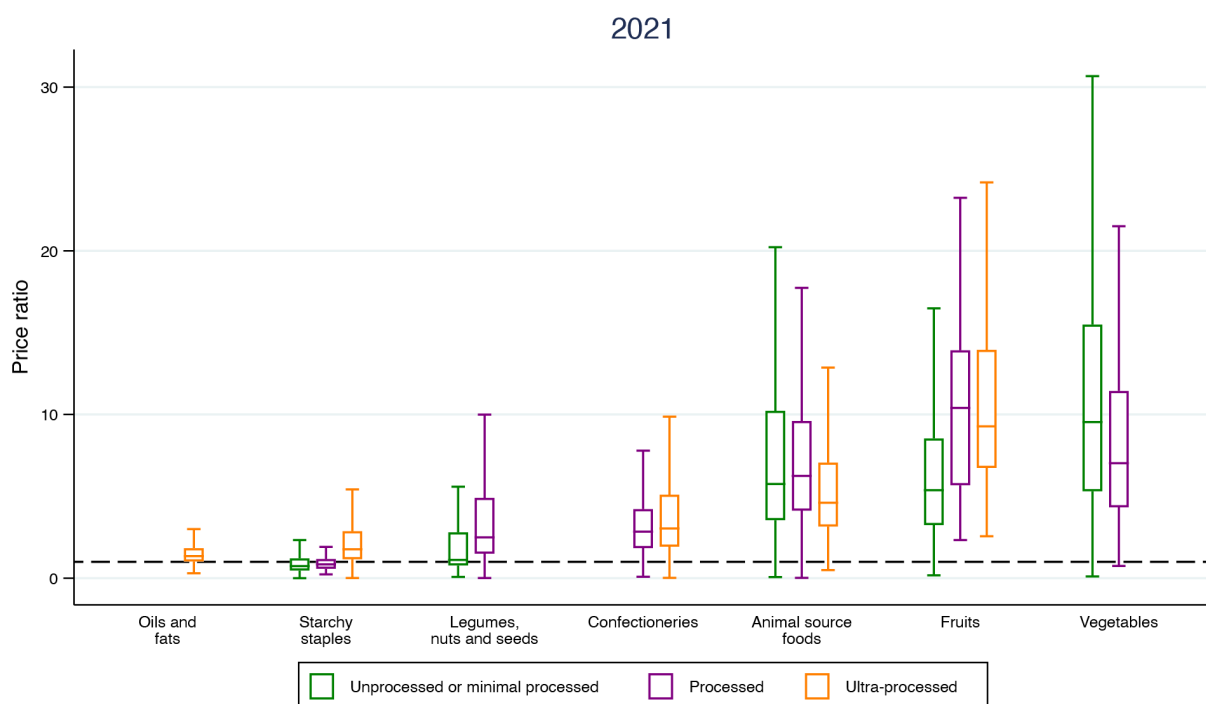
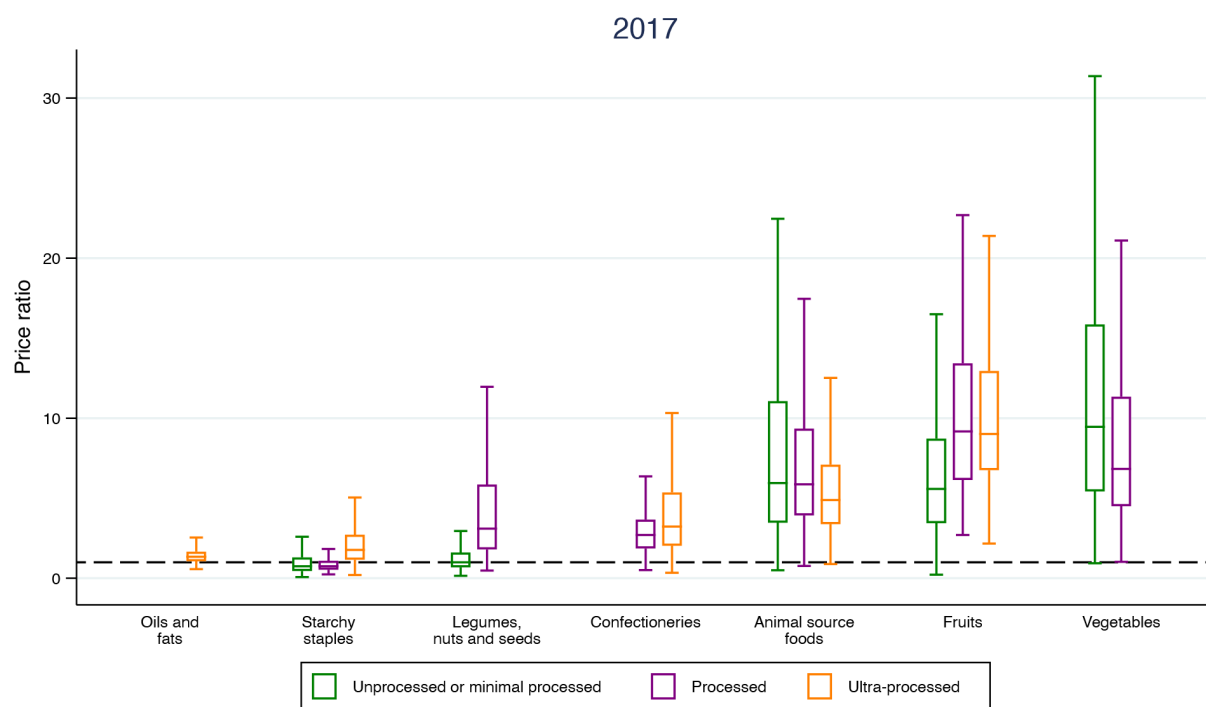
Source: Authors' own elaboration.

These comparisons differ considerably when comparing the prices of items in each food group by processing level. When the analysis controls for food groups, ultra-processed options are statistically significantly more expensive than unprocessed or minimally processed options (Table A2). Among starchy staples, ultra-processed options are significantly more expensive than unprocessed or processed options. In 2021, ultra-processed starchy staples were 2.2 times as expensive as unprocessed or minimally processed starchy staples, while processed starchy staples were the same prices as unprocessed or minimally processed starchy staples on average. Similarly, among sweets, ultra-processed options are significantly more expensive

than processed options. In 2021, ultra-processed sweets were 4.2 times as expensive as basic starchy staples, while processed sweets were 3.5 times as expensive. However, ultra-processed starchy staples and sweets are much less expensive than unprocessed or minimally processed vegetables, fruits and animal-source foods (fresh and dried items), which make up most (72.4 percent) of the unprocessed or minimally processed foods category. These findings are also consistent across all three rounds of data observed (Figure 4 and Table A4). This result is driven by the food group composition of each processing category. Ultra-processed animal-source foods are less expensive, likely due to the inclusion of lower cost ingredients from other food groups, especially sugars, oils and flours or other starches, in addition to dairy, fish, or meat. Ultra-processed foods are likely to be categorized in lower-cost food groups, while unprocessed or minimally processed items are likely to be in higher-cost food groups. Most ultra-processed foods observed were starchy staples and sweets (36.4 and 41.7 percent of ultra-processed food items observed, respectively). In contrast, the food groups that had the highest price premiums in Figure 1 contribute fewer ultra-processed items: 15.9 percent of all ultra-processed items were animal-source foods and 2.7 percent were fruits, while no ultra-processed vegetables were present in the data.

Figure 4. Ratio of food prices relative to the price of basic starchy staples, by processing level and year





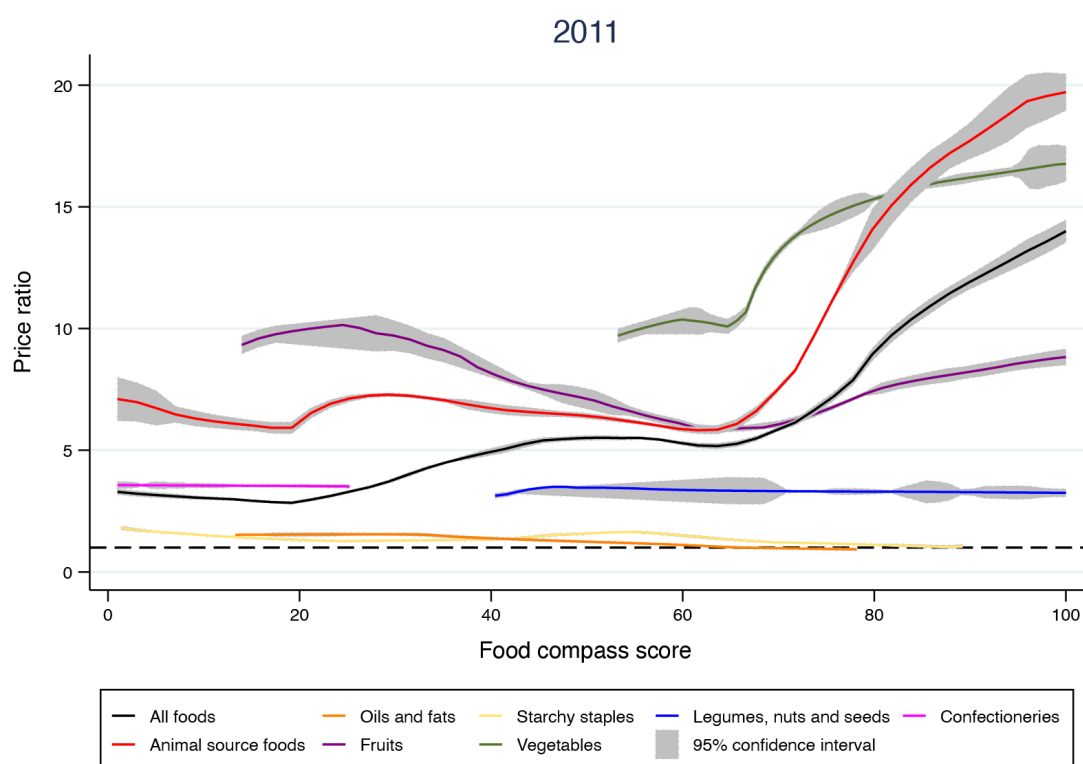
Notes: Data shown are price ratios (price per kilocalorie of edible matter for each food item in each country/year divided by the average price per kilocalorie of basic starchy staples in that country/year). Basic starchy staples are those classified as NOVA 1 (unprocessed or minimally processed foods). Prices are national average retail food prices from ICP in 2011, 2017 and 2021. For each box plot, the centre line indicates the median; the box shows interquartile range (25th to 75th percentile); and the outer lines show the range of data points that fall within 1.5*IQR around the box. Observations that fall outside of 1.5*IQR are not shown. The dashed line shows where the price ratio is equal to 1.

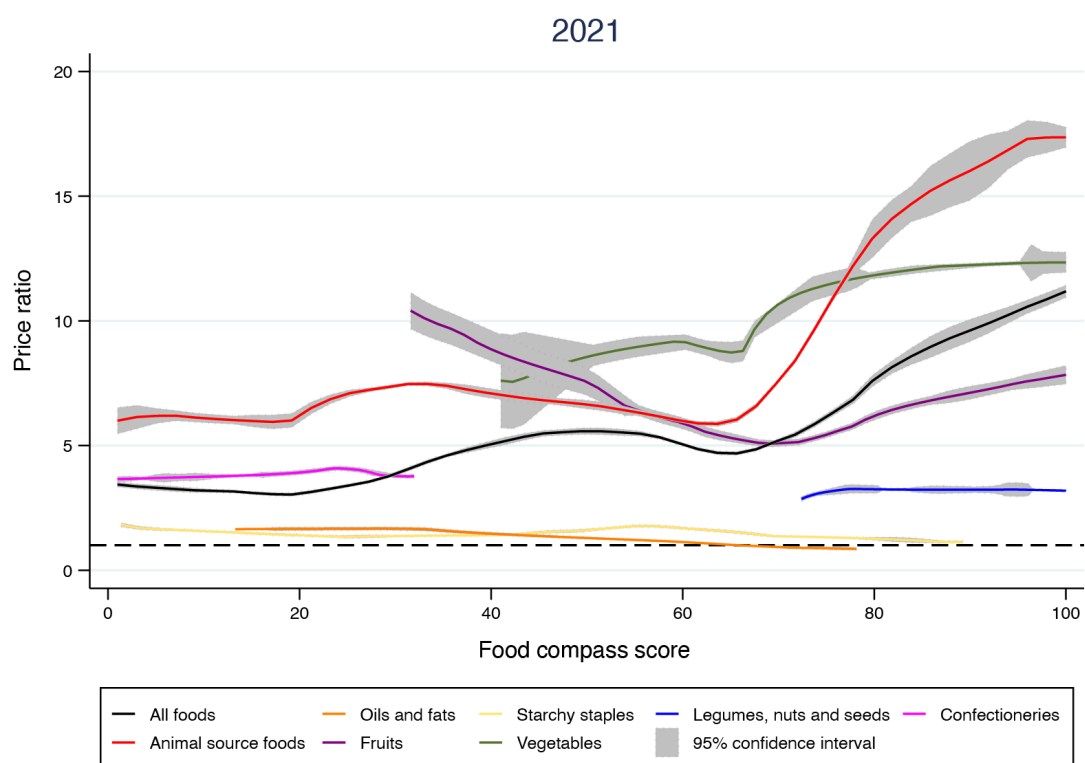
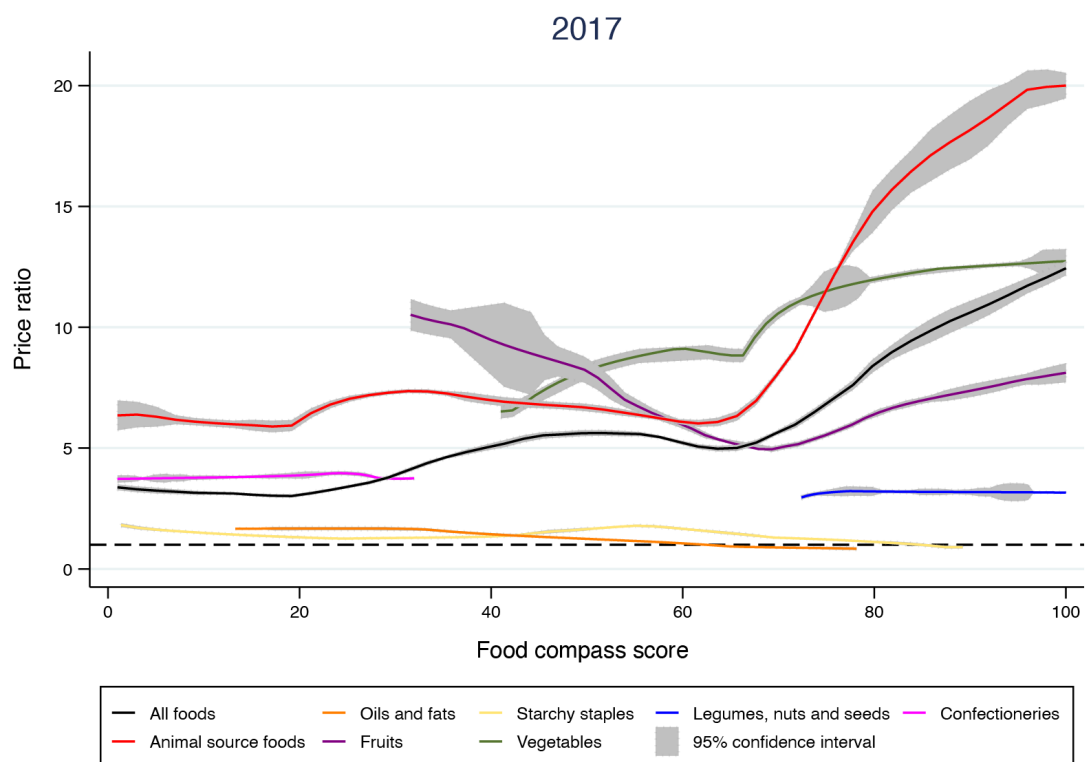
Source: Authors' own elaboration.

4.1.3 Price variation by nutritional quality

Healthier foods are generally more expensive per kilocalorie, though this relationship is not linear. The most nutritious foods, defined as those with FCS between 80 and 100, are much more expensive than less nutritious foods on average (Figure 5, black line). In 2021, foods with the highest FCS (80–100) were 7.5 times as expensive as foods with the lowest FCS (≤ 20) (Table A3). However, considering trends across all food groups masks diverse within-group trends. Within each food group, healthier options are not necessarily more expensive. Among animal-source foods and vegetables, the most nutritious options are much more expensive than less nutritious options on average (Figure 5, red and green lines). In contrast, among starchy staples, oils and fats, legumes, nuts and seeds, and sweets, price varies very little with nutritional quality (Figure 5, yellow, orange, pink and blue lines). While most fruits are highly nutritious, the most expensive options – such as tinned fruits in syrup – have a lower nutrient profile than other fruits (Figure 5, purple line). As with comparisons between food groups and by processing level, the same trends in price by nutritional quality are observed in 2021, 2017 and 2011 (Figure 5 and Table A3).

Figure 5. Ratio of food prices relative to the price of basic starchy staples, by nutritional profile, food group and year





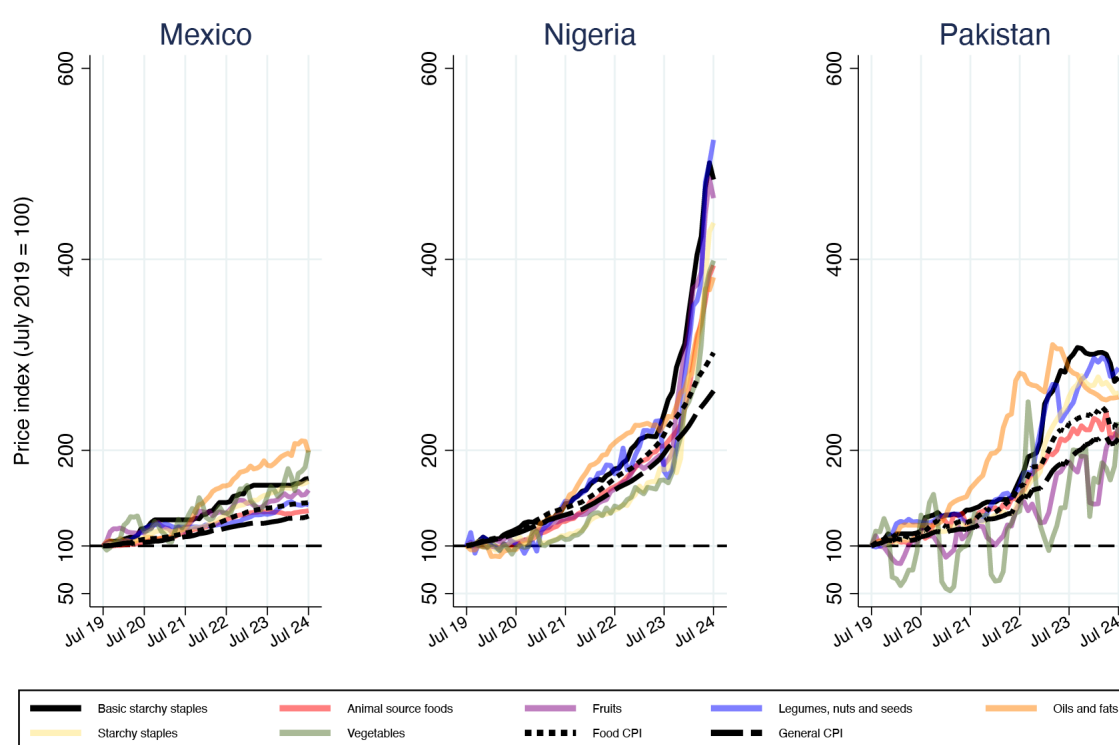
Notes: Data shown are price ratios (price per kilocalorie of edible matter for each food item in each country/year divided by the average price per kilocalorie of basic starchy staples in that country/year). Basic starchy staples are those classified as NOVA 1 (unprocessed or minimally processed foods). Prices are national average retail food prices from ICP in 2011, 2017 and 2021. Lines show local polynomial smooths with 95 percent confidence intervals in grey. The dashed line shows where the price ratio is equal to 1.

Source: Authors' own elaboration.

4.2 Country-level results

During the post-COVID-19 pandemic period, price increases were rapid across many product categories and food price inflation was higher than overall inflation as shown in Figure 6. While average price increases are not directly comparable with the food CPI, in which foods are weighted by expenditure shares, Figure 6 also shows that price increases for less expensive food groups were higher than food inflation more broadly. Price increases were especially high in Nigeria and Pakistan, but price increases for oils and fats were unusually large across all three countries in 2021 and 2022. Prices for oils and fats remained elevated above all other food groups in Pakistan until well into 2023, and in Mexico until the middle of 2024. In Nigeria, prices for legumes, nuts and seed also rose sharply during this period, while rising prices for all food groups outpaced both general and food inflation in 2024. Prices for basic starchy staples rose at a faster rate than food price inflation at different times in all three countries: in Nigeria, between 2021 and 2022; in Pakistan, between 2022 and 2023; and in Mexico, between 2020 and 2021, then again in 2022 and 2023. While price increases affected different food groups at different times during the study period, these patterns indicate that the overall impact of inflation in the post-pandemic period may have been especially severe for starchy staples, oils and fats, and legumes, nuts and seeds, and comparatively less severe for animal-source foods and fruits.

Figure 6. National inflation dynamics and nominal price trends over time

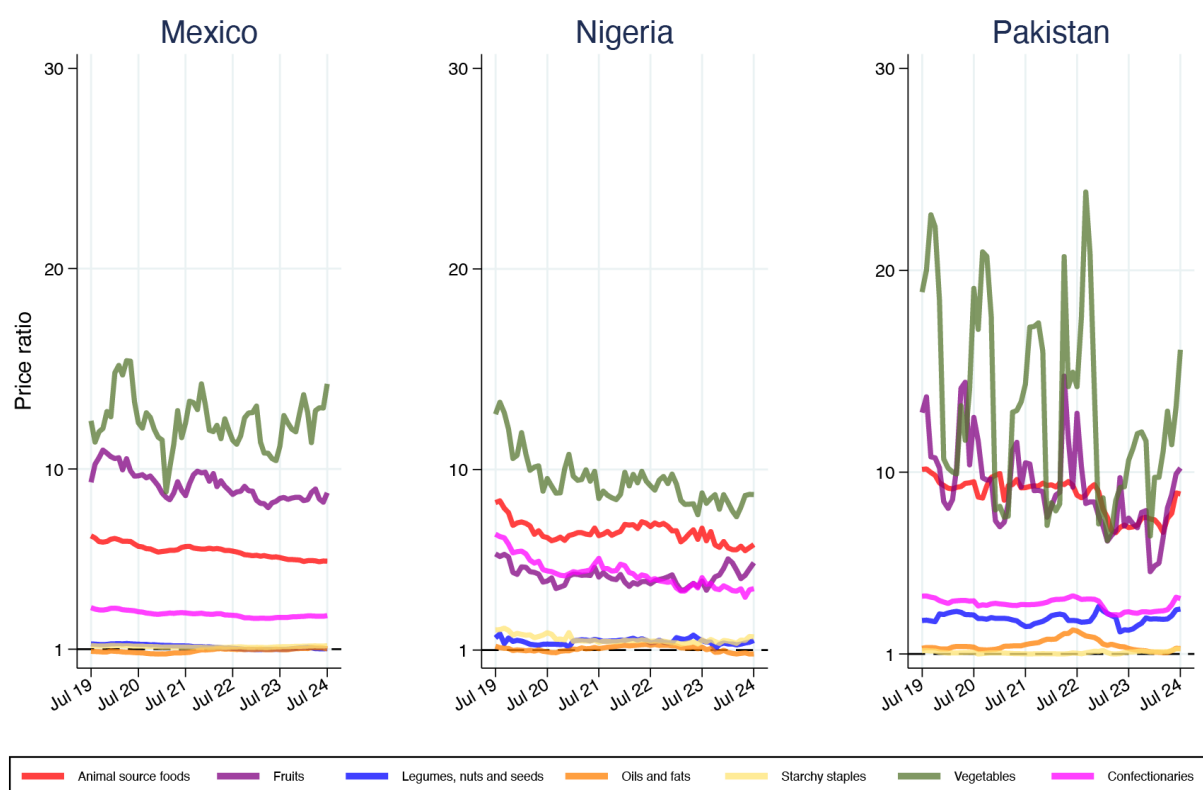


Notes: CPI = consumer price index. Data shown are overall CPI, food CPI and median nominal prices by food group in each country. Basic starchy staples are those classified as NOVA 1 (unprocessed or minimally processed foods). The category “Basic starchy staples” includes unprocessed or minimally processed foods (NOVA 1). Prices are expressed in nominal local currency units per kilocalorie of edible matter and indexed to the initial period (July 2019=100). For Mexico and Nigeria, CPI is shown at the national level as reported; for Pakistan, CPI shown is the average of reported urban and rural CPI.

Source: Authors’ own elaboration.

Comparing average prices to basic starchy staples by food group reveals some patterns that persist across all three countries. Figure 7 depicts trends over time in the ratio of average prices by food group (dashed coloured lines) relative to the average price for basic starchy staples (dashed horizontal black line). The height of each coloured line above the dashed black line indicates the premium to purchase foods other than unprocessed or minimally processed staples, such as rice, wheat grain or flour, tubers, or noodles. The price premium above basic starchy staples is consistently higher for vegetables than for all other food groups, closely followed by fruits and animal-source foods. These foods also exhibit a higher degree of seasonal volatility, with consistent fluctuations in price premium that are both larger and more frequent than comparable spikes in the less perishable food groups. This fundamental pattern of more expensive perishable foods remains unchanged across the time series and across all food groups, but the price premium for these items declined substantially in Nigeria during the post-pandemic inflationary period. In Mexico, price premiums for fruits and animal-source foods declined modestly relative to basic starchy staples during this time period.

Figure 7. Real price premiums by food group relative to basic starchy staples

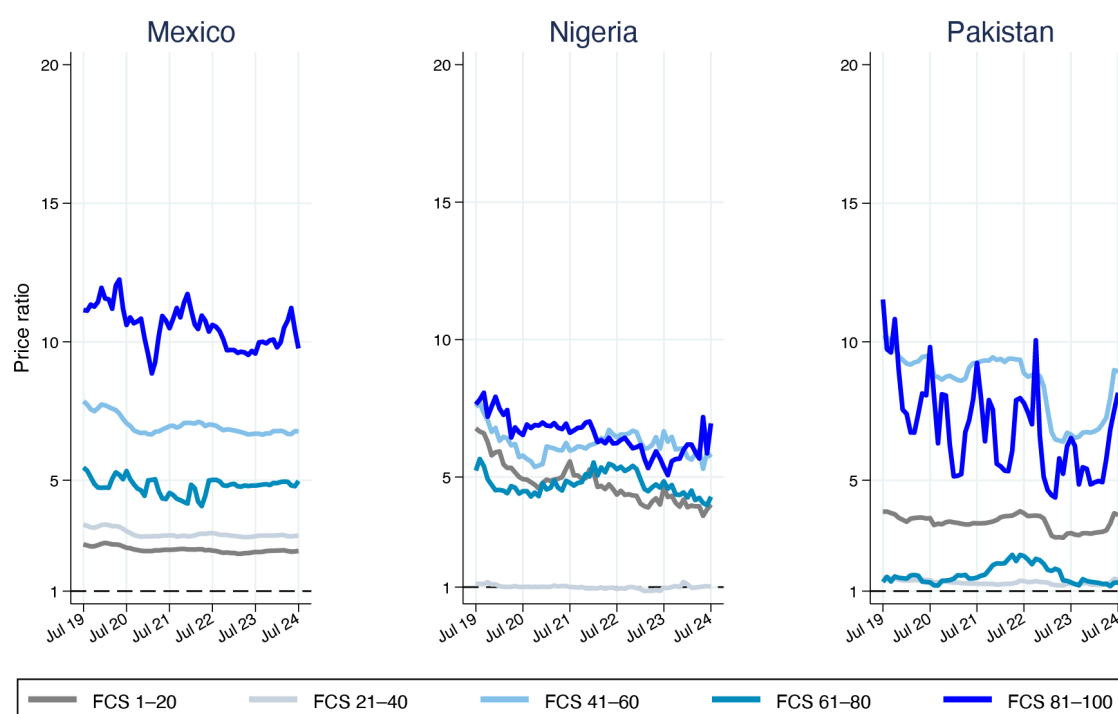


Notes: Data shown are median real food prices for each food group divided by the median food prices across all unprocessed or minimally processed starchy staples (NOVA 1) in each country. Prices are expressed in inflation-adjusted July 2019 local currency units per kilocalorie of edible matter and converted to three-month rolling mean before food group medians are calculated. Dashed black lines indicate where the price ratio is equal to 1.

Source: Authors' own elaboration.

Price premiums for foods with higher nutrient profiles are not systematically different, but specific country-level trends suggest some relationship with healthfulness (Figure 8). In all three countries, foods with the highest FCS (80–100) are substantially more expensive than basic starchy staples. However, in all three countries, the price premium for items with the lowest FCS (≤ 20) is still larger than the price premium for some items with higher scores. In Mexico, the differences between categories are large and well-defined, partly due to an extensive food list (Table A4). The pattern is also relatively stable over time with the exception of the highest-scoring category of foods, whose average price declined markedly over the analysis period. In Nigeria, foods across the spectrum of FCS are clustered in a narrower range of prices relative to basic starchy staples, and prices for the highest-scoring two categories of FCS slightly declined over the analysis period. The highest-scoring items across all food groups are consistently more expensive than the least healthy items. However, foods with moderately low FCS scores (including rice, maize products and margarine) are substantially less expensive than the lowest-scoring foods (including sweets, processed meats and instant noodles). In Pakistan, the highest-scoring items across all food groups have substantially higher price premiums relative to the lowest-scoring items. Foods with the second-highest range of FCS (60–80), including wheat flour, semolina, vegetable oil, taro and cauliflower) have a smaller price premium than the lowest-scoring items. Meanwhile, the highest FCS category is similar in price to the middle FCS category, while the second-highest category is comparatively close to the cost of basic starchy staples.

Figure 8. Real price premiums by Food Compass Score category relative to basic starchy staples

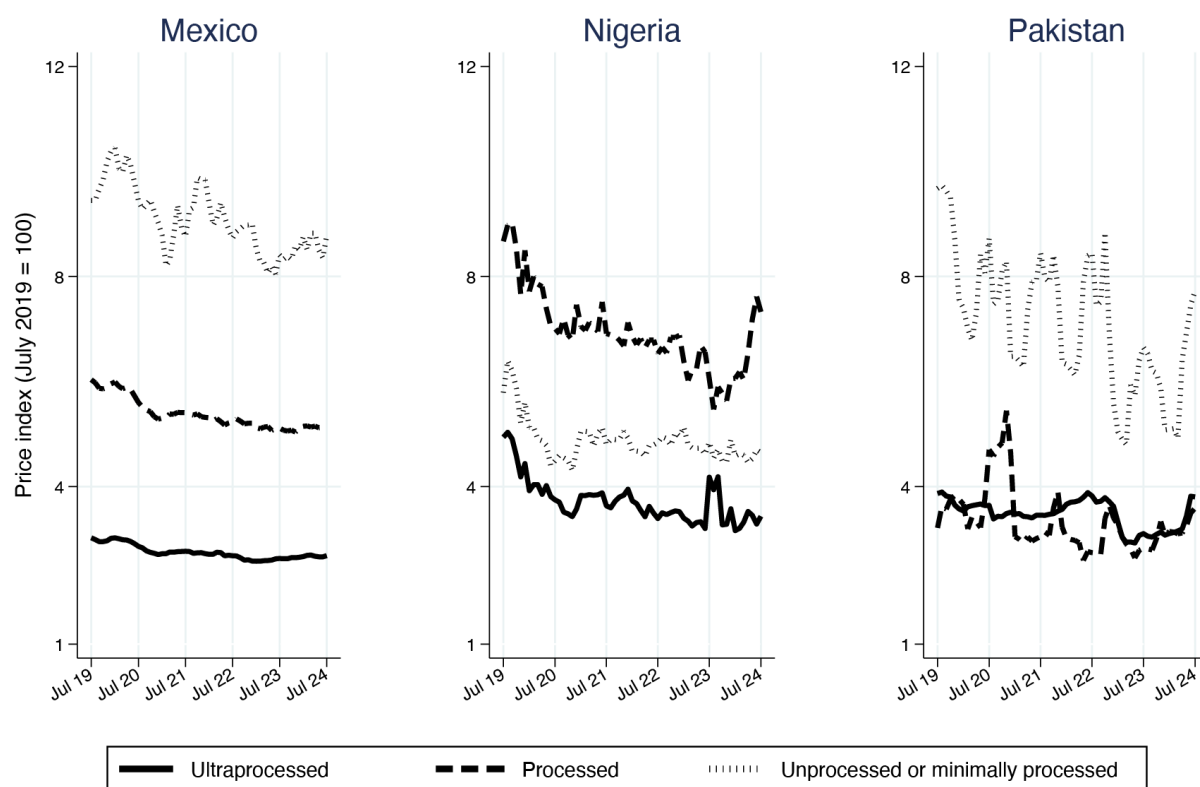


Notes: FCS = Food Compass Score. Data shown are median real food prices for Food Compass Score category within each food group, divided by the median real food price across all unprocessed or minimally processed starchy staples (NOVA 1). Prices are expressed in inflation-adjusted July 2019 local currency units per kilocalorie of edible matter and converted to three-month rolling mean before food group medians are calculated. Dashed lines indicate where the price ratio is equal to 1.

Source: Authors' own elaboration.

Price premiums for ultra-processed foods tend to be lower than those for unprocessed foods when items from all food groups are compared, but unprocessed or minimally processed foods are not necessarily the most expensive (Figure 9). In Nigeria, processed foods were the most expensive relative to basic starchy staples. Across the entire study period, the median price premium in Nigeria for processed foods over basic starchy staples was 6.9, compared to only 3.9 for ultra-processed foods. However, the price premiums for all three NOVA categories declined substantially during the analysis period. In Mexico and Pakistan, unprocessed or minimally processed foods were consistently the most expensive options throughout the time period. In Pakistan, price premiums for unprocessed or minimally processed foods were 7.4, nearly twice as large as the premiums for processed and ultra-processed foods. In Mexico, differences in price premiums across processing categories were large and well-defined, with premiums for unprocessed or minimally processed foods averaging about 9.0, for processed foods about 5.3 and for ultra-processed foods about 2.7.

Figure 9. Real price premiums by NOVA processing category relative to basic starchy staples

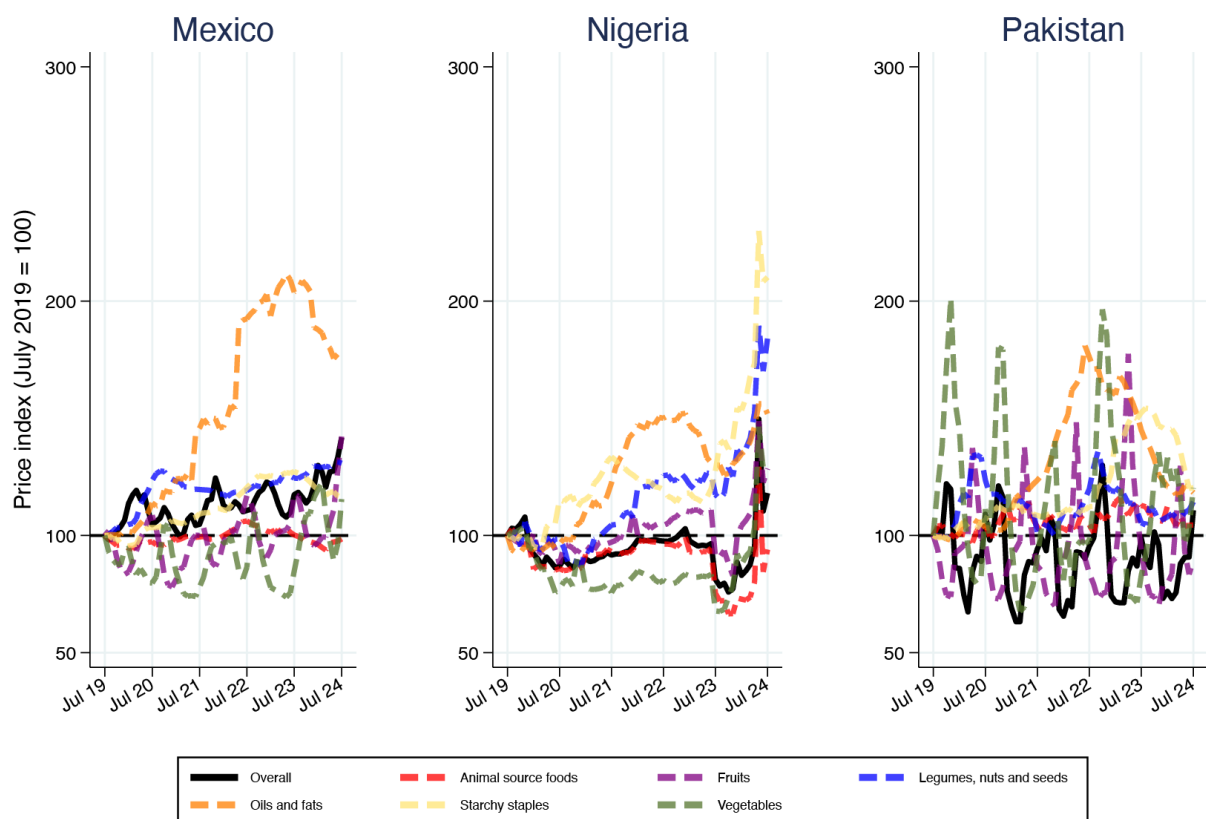


Notes: Data shown are median food prices for NOVA processing category within each food group, divided by the median food price across all unprocessed or minimally processed starchy staples (NOVA 1). Prices are expressed in inflation-adjusted July 2019 local currency units per kilocalorie of edible matter and converted to three-month rolling mean before food group medians are calculated. Dashed lines indicate where the price ratio is equal to 1.

Source: Authors' own elaboration.

Price dynamics for least-cost foods differed substantially from price dynamics across the full price range. This resulted in different implications for least-cost healthy diets in each of the three countries, as shown in Figure 10, which shows price indices constructed as the weighted average of items included in least-cost diets. In Nigeria, the average price of all least-cost items remained at or below the value in the reference period until early 2024. In Pakistan, the average price exhibited more volatility but oscillated around the reference value through 2024. In Mexico, the average price rose steadily over time, reaching a level that was 41.6 percent higher than baseline by July 2024. This indicates that while price increases across all items within food groups were severe (as shown in Figure 6), the average price of least-cost healthy items rose more slowly (as in Mexico and Nigeria) or not at all (as in Pakistan).

Figure 10. Weighted-average price index of items included in least-cost healthy diets



Notes: Data shown are average real food prices for each food weighted by the frequency of inclusion in least-cost healthy diets between July 2019 and July 2024 and indexed to July 2019. Item inclusion reflects only foods recommended as part of a healthy diet and exclude items such as cured meats, snack foods and foods with added sugar. Prices are expressed in inflation-adjusted July 2019 local currency units per kilocalorie of edible matter and converted to three-month rolling means before food group averages are calculated. Food group trends are weighted by frequency of inclusion at the food group level, and overall trends are weighted by frequency of inclusion across all food groups.

Source: Authors' own elaboration.

5. Discussion and conclusion

These analyses show that the food group to which a retail food item belongs is the primary predictor of its relative price, rather than level of processing or nutrient profiling score. This finding is consistent both in our global analysis and in the three country case studies presented here. Across the results, vegetables, fruits and animal-source foods have high price premiums compared to basic starchy staples. In the global analysis, we additionally find that the correlation between food group and processing level explains much of the apparent low cost of ultra-processed foods, as most ultra-processed foods are inexpensive starchy staples and sweets. Meanwhile, in Mexico, Nigeria and Pakistan, while the healthiest foods available in the marketplace are commonly more expensive than the least healthy items, there is not a consistent gradient between price and healthfulness. In some cases, foods with very high nutrient profile are substantially less expensive than lower-scoring options. At the same time, there is an ambiguous connection between food processing and healthfulness. High relative prices of perishable foods make these foods less affordable than more shelf-stable ultra-processed options. In other cases, high relative prices for certain processed and ultra-processed foods may be an important constraint on access to healthy diets, as these foods are often labour-saving options that are preferable to unprocessed foods that can require substantial investments of time and fuel to prepare.

The global analysis reveals consistent differences in retail food prices between food groups and by processing level and nutritional quality. There are no meaningful differences in these trends by national income level or between retail price observations from 2011, 2017 and 2021. This suggests that these differences are a consistent feature of food markets globally, even when and where retail prices are higher or lower overall. It is often hypothesized that consumption of unhealthy diets is driven by a lack of inexpensive, healthy foods at retail markets around the world. If this were the case, we would expect foods with lower nutritional profile and ultra-processed foods to be systematically less expensive, and few nutritious and inexpensive foods to be available. These trends would be especially apparent in higher-income countries where consumption of less healthful foods is highest.

We do observe that the most healthful foods are more expensive than the least healthful foods available. However, these trends are not consistent within food groups. In addition, we do observe that very high-cost, nutritious foods are available in nearly all countries, which may drive the perception that healthful foods are expensive. However, low-cost, nutritious options are also available. Similarly, we observe that ultra-processed foods are less expensive per unit of dietary energy compared to unprocessed foods when averaging across all retail foods. However, these trends do not hold within food groups. In fact, ultra-processed starchy staples – which are often preferred by consumers over more nutrient dense grains – are much more expensive per kilocalorie compared to unprocessed or minimally processed starchy staples.

Each analysis is subject to several important limitations. The global analysis incorporates cross-sectional data, and each of the three ICP data collection rounds represents slightly different food lists. The three countries analysed here are case studies and are not representative of their geographic regions, national income level, or of other countries elsewhere in the world. Each country experienced distinctive economic and political events during this period that would have contributed to inflationary dynamics. In Nigeria, for example, these events included a currency redesign and subsequent devaluation as well as the removal of long-standing fuel subsidies. Pakistan experienced climatic shocks during this period.

Findings from these case studies should be interpreted cautiously in the context of their regions and the rest of the world.

Taken together, the global and country-level results indicate that, though many nutritious and under-consumed foods are more expensive than basic starchy staples, other ways of categorizing healthy and unhealthy foods do not reveal systematic price differences. While there may exist important differences in price within food groups and between different nutritional attributes of foods at specific times and in specific places, these are not facts that can be generalized to food systems around the world. In contrast, the price premium to purchase animal-source foods, vegetables and fruits is likely a fundamental feature of these foods, which tend to be highly perishable, require high inputs of labour, and are less easily stored than other food groups.

These findings have important food security implications that may not be especially surprising. For the roughly one-third of people who cannot afford to buy the foods required for health, relative prices between food categories influence food choice in that energy needs must be met within the household's budget constraint. For the remaining two-thirds of people around the globe who could choose a healthy diet but on average do not, relative food prices are likely to provide a weak signal. Food prices may provide an even weaker signal for consumer decision-making compared to the many messages from marketing, advertising and other elements of the food environment and broader cultural factors directing food choice toward less healthy but aspirational items.

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Annex

Table A1. Price premiums relative to basic starchy staples worldwide, by food group and national income level, 2021

	(1)	(2)	(3)	(4) country FE
Oils and fats	0.00	0.00	0.00	0.00
Starchy staples	0.23	0.21	0.20	0.19
Legumes, nuts and seeds	1.89*	1.86*	2.49*	2.41*
Sweets	2.45*	2.41*	2.71*	2.60*
Animal-source foods	7.86*	7.88*	6.79*	6.73*
Fruits	5.95*	5.94*	5.46*	5.41*
Vegetables	10.61*	10.58*	7.58*	7.50*
Low-income countries (LICs)		0.00	0.00	0.00
Lower-middle-income countries (LMICs)		1.06*	0.16	0.15
Upper-middle-income countries (UMICs)		0.88*	0.03	-0.17
High-income countries (HICs)		0.35*	-0.28	-0.33
Starchy staples x LMICs			-0.07	-0.19
Starchy staples x UMICs			-0.07	-0.09
Starchy staples x HICs			0.22	0.21
Legumes, nuts and seeds x LMICs			-0.52	-0.60
Legumes, nuts and seeds x UMICs			-0.40	-0.44
Legumes, nuts and seeds x HICs			-1.14	-1.08
Sweets x LMICs			-0.06	-0.11
Sweets x UMICs			-0.15	-0.14
Sweets x HICs			-0.70	-0.59
Animal-source foods x LMICs			1.48*	1.51*
Animal-source foods x UMICs			1.44	1.37
Animal-source foods x HICs			1.03	1.05
Fruits x LMICs			0.98	1.00
Fruits x UMICs			0.68	0.75
Fruits x HICs			0.10	0.14
Vegetables x LMICs			3.38*	3.34*
Vegetables x UMICs			3.36*	3.43*
Vegetables x HICs			3.69*	3.80*
Intercept	1.24*	0.62*	1.28*	1.29*

Notes: Results shown are regression coefficients of price ratio (price per kilocalorie of edible matter for each food item in each country/year divided by the average price per kilocalorie of basic starchy staples in that country/year) on food group and national income level. Basic starchy staples are those classified as NOVA 1 (unprocessed or minimally processed foods). Prices are national average retail food prices from ICP in 2021. * Coefficients that are significant at the 0.05 level.

Source: Authors' own elaboration.

Table A2. Price premiums relative to basic starchy staples worldwide, by food group and processing level, 2021

	(1)	(2)	(3)	(4) country FE
Unprocessed	0.00	0.00	0.00	0.00
Processed	0.35*	0.22	0.40	0.34
Ultra-processed	-3.38*	-0.18	3.85*	3.98*
Oils and fats		0.00	0.00	0.00
Starchy staples		-0.25	3.23*	3.28*
Legumes, nuts and seeds		1.21	5.12*	5.13*
Sweets		2.35*	2.62*	2.45*
Animal-source foods		7.27*	11.80*	11.82*
Fruits		5.40*	8.96*	9.05*
Vegetables		9.94*	13.95*	14.04*
Unprocessed x oils and fats			0.00	0.00
Unprocessed x sweets			0.00	0.00
Processed x oils and fats			0.00	0.00
Processed x starchy staples			-0.37	0.09
Processed x legumes, nuts and seeds			0.08	0.17
Processed x sweets			2.68*	2.89*
Processed x animal-source foods			-0.61	-0.38
Processed x fruits			3.44*	3.33*
Processed x vegetables			0.00	0.00
Ultra-processed x starchy staples			-2.62*	-2.74*
Ultra-processed x legumes, nuts and seeds			0.00	0.00
Ultra-processed x sweets			0.00	0.00
Ultra-processed x animal-source foods			-7.86*	-7.99*
Ultra-processed x fruits			0.00	0.00
Ultra-processed x vegetables			0.00	0.00
Intercept	7.18*	1.80*	-2.23*	-2.43*

Notes: Results shown are regression coefficients of price ratio (price per kilocalorie of edible matter for each food item in each country/year divided by the average price per kilocalorie of basic starchy staples in that country/year) on food group and processing level. Basic starchy staples are those classified as NOVA 1 (unprocessed or minimally processed foods). Prices are national average retail food prices from ICP in 2021. * Coefficients that are significant at the 0.05 level.

Source: Authors' own elaboration.

Table A3. Price premiums relative to basic starchy staples worldwide, by food group and nutrient profile, 2021

	(1)	(2)	(3)	(4) country FE
FCS $\leq$ 20	0.00	0.00	0.00	0.00
20 < FCS $\leq$ 40	-0.75*	0.29	1.31	1.13
40 < FCS $\leq$ 60	2.76*	-0.07	6.68*	6.69*
60 < FCS $\leq$ 80	0.01	0.07	0.32	0.12
80 < FCS $\leq$ 100	7.48*	6.48*	11.25*	11.16*
Oils and fats		0.00	0.00	0.00
Starchy staples		-0.08	1.29	1.03
Legumes, nuts and seeds		-3.47*	-8.72*	-8.94*
Sweets		2.60*	3.15*	2.84*
Animal-source foods		6.36*	5.30*	5.13*
Fruits		1.74*	-4.35*	-4.49*
Vegetables		5.17*	0.38	0.26
FCS $\leq$ 20 x legumes, nuts and seeds			0.00	0.00
FCS $\leq$ 20 x fruits			0.00	0.00
FCS $\leq$ 20 x vegetables			0.00	0.00
20 < FCS $\leq$ 40 x starchy staples			-1.87*	-1.69
20 < FCS $\leq$ 40 x legumes, nuts and seeds			0.00	0.00
20 < FCS $\leq$ 40 x sweets			-0.70	-0.63
20 < FCS $\leq$ 40 x animal-source foods			-1.06	-0.94
20 < FCS $\leq$ 40 x fruits			13.09*	13.30*
20 < FCS $\leq$ 40 x vegetables			0.00	0.00
40 < FCS $\leq$ 60 x oils and fats			0.00	0.00
40 < FCS $\leq$ 60 x starchy staples			-6.61*	-6.54*
40 < FCS $\leq$ 60 x legumes, nuts and seeds			0.00	0.00
40 < FCS $\leq$ 60 x sweets			0.00	0.00
40 < FCS $\leq$ 60 x animal-source foods			-5.71*	-5.80*
40 < FCS $\leq$ 60 x fruits			4.11*	4.02*
40 < FCS $\leq$ 60 x vegetables			0.00	0.00
60 < FCS $\leq$ 80 x starchy staples			-0.97	-0.89
60 < FCS $\leq$ 80 x legumes, nuts and seeds			11.49*	11.61*
60 < FCS $\leq$ 80 x sweets			0.00	0.00
60 < FCS $\leq$ 80 x animal-source foods			-0.99	-0.91
60 < FCS $\leq$ 80 x fruits			7.95*	8.04*
60 < FCS $\leq$ 80 x vegetables			10.99*	10.62*
80 < FCS $\leq$ 100 x oils and fats			0.00	0.00
80 < FCS $\leq$ 100 x starchy staples			-11.22*	-10.95*
80 < FCS $\leq$ 100 x legumes, nuts and seeds			0.00	0.00

	(1)	(2)	(3)	(4) country FE
80 < FCS ≤ 100 x sweets			0.00	0.00
80 < FCS ≤ 100 x animal-source foods			0.00	0.00
80 < FCS ≤ 100 x fruits			0.00	0.00
80 < FCS ≤ 100 x vegetables			0.00	0.00
Intercept	3.42*	1.09*	0.52	0.65

Notes: FCS = Food Compass Score. Results shown are regression coefficients of price ratio (price per kilocalorie of edible matter for each food item in each country/year divided by the average price per kilocalorie of basic starchy staples in that country/year) on food group and FCS. Basic starchy staples are those classified as NOVA 1 (unprocessed or minimally processed foods). Prices are national average retail food prices from ICP in 2021. FCS is categorized in quintiles; low FCS denotes less healthful foods; high FCS denotes more healthful foods. * Coefficients that are significant at the 0.05 level.

Source: Authors' own elaboration.

Table A4. Price ratios by food group, income group, NOVA category and Food Compass Score quintile, 2021

	N	Mean	Median	Min	Max
Food group					
Oils and fats	1 434	1.24	0.98	0.02	5.84
Starchy staples	4 932	1.48	1.05	0.01	16.14
Legumes, nuts and seeds	1 120	3.13	1.81	0.01	32.06
Sweets	3 439	3.69	2.75	0.02	44.26
Animal-source foods	7 996	9.10	5.75	0.02	291.96
Fruits	2 257	7.19	5.79	0.18	67.40
Vegetables	2 294	11.85	9.17	0.11	81.09
Income group					
Low-income countries					
Oils and fats	234	1.28	0.89	0.18	5.09
Starchy staples	677	1.48	1.02	0.01	7.22
Legumes, nuts and seeds	149	3.77	2.04	0.01	23.76
Sweets	429	3.99	3.12	0.02	29.77
Animal-source foods	1 277	8.07	6.12	0.02	79.74
Fruits	325	6.74	5.41	0.25	34.24
Vegetables	301	8.86	7.11	0.11	53.19
Lower-middle-income countries					
Oils and fats	413	1.44	1.03	0.02	5.46
Starchy staples	1 447	1.58	1.05	0.01	16.01
Legumes, nuts and seeds	335	3.41	1.84	0.08	20.70
Sweets	1 105	4.10	3.18	0.07	44.26
Animal-source foods	2 457	9.71	6.35	0.13	201.48
Fruits	657	7.88	6.32	0.18	67.40
Vegetables	642	12.40	8.91	0.25	66.51
Upper-middle-income countries					
Oils and fats	336	1.31	1.11	0.18	4.83
Starchy staples	1 237	1.44	1.02	0.06	11.73
Legumes, nuts and seeds	301	3.40	2.13	0.42	32.06
Sweets	850	3.87	2.84	0.06	39.23
Animal-source foods	1 910	9.54	5.41	0.79	291.96
Fruits	600	7.44	5.94	0.43	50.11
Vegetables	602	12.25	9.04	0.63	81.09
High-income countries					
Oils and fats	431	1.00	0.94	0.10	5.84
Starchy staples	1 508	1.42	1.09	0.10	16.14
Legumes, nuts and seeds	318	2.35	1.50	0.32	23.23
Sweets	1 011	3.01	2.33	0.13	34.06
Animal-source foods	2 262	8.81	4.96	0.50	141.88

	N	Mean	Median	Min	Max
Fruits	647	6.55	5.52	0.48	30.62
Vegetables	713	12.27	10.68	0.76	61.55
NOVA category					
Unprocessed	12 754	7.18	4.30	0.01	213.35
Processed	3 519	7.53	4.93	0.01	291.96
Ultra-processed	5 235	3.80	2.67	0.01	44.26
Food group					
Oils and fats					
Ultra-processed	175	1.61	1.35	0.31	5.84
Starchy staples					
Unprocessed	2 946	1.00	0.75	0.01	6.79
Processed	82	1.02	0.85	0.24	5.85
Ultra-processed	1 904	2.23	1.77	0.01	16.14
Legumes, nuts and seeds					
Unprocessed	574	2.89	1.12	0.08	32.06
Processed	464	3.36	2.45	0.01	19.50
Sweets					
Processed	669	3.47	2.84	0.09	37.60
Ultra-processed	2 184	4.24	3.03	0.02	44.26
Animal-source foods					
Unprocessed	5 276	9.57	5.75	0.07	213.35
Processed	1 868	9.35	6.23	0.02	291.96
Ultra-processed	830	5.55	4.61	0.50	37.72
Fruits					
Unprocessed	1 980	6.72	5.37	0.18	67.40
Processed	135	10.56	10.40	2.33	50.11
Ultra-processed	142	10.57	9.19	2.57	40.86
Vegetables					
Unprocessed	1 978	11.72	9.55	0.11	81.09
Processed	301	12.11	7.03	0.76	66.51
Food Compass Score (FCS)					
FCS $\leq$ 20	4 969	3.42	2.46	0.01	44.26
20 < FCS $\leq$ 40	3 972	2.67	1.37	0.01	40.86
40 < FCS $\leq$ 60	4 919	6.18	4.58	0.01	61.99
60 < FCS $\leq$ 80	2 864	3.43	2.32	0.01	42.91
80 < FCS $\leq$ 100	6 748	10.90	6.97	0.01	291.96
Food Compass Score					
FCS $\leq$ 20					
Oils and fats	97	0.52	0.42	0.17	2.21
Starchy staples	1 049	1.81	1.44	0.01	9.81
Sweets	3 353	3.68	2.70	0.02	44.26

	N	Mean	Median	Min	Max
Animal-source foods	470	5.82	4.53	0.44	37.72
20 < FCS ≤ 40					
Oils and fats	616	1.83	1.65	0.02	5.84
Starchy staples	2 359	1.25	0.79	0.01	16.14
Sweets	86	4.29	4.19	0.53	10.09
Animal-source foods	769	6.06	5.20	0.03	35.95
Fruits	142	10.57	9.19	2.57	40.86
40 < FCS ≤ 60					
Starchy staples	651	1.89	1.53	0.01	9.08
Animal-source foods	3 794	6.79	5.20	0.02	61.99
Fruits	320	6.97	5.63	0.18	50.11
Vegetables	154	7.59	5.20	0.76	61.90
60 < FCS ≤ 80					
Oils and fats	721	0.84	0.57	0.02	5.46
Starchy staples	618	1.16	0.71	0.06	5.48
Legumes, nuts and seeds	168	3.60	2.99	0.01	19.50
Animal-source foods	922	5.15	4.40	0.42	34.26
Fruits	278	4.45	3.92	0.82	20.70
Vegetables	157	12.21	7.96	1.58	42.91
80 < FCS ≤ 100					
Starchy staples	255	1.84	1.49	0.01	6.22
Legumes, nuts and seeds	952	3.05	1.66	0.03	32.06
Animal-source foods	2 041	17.07	9.64	0.09	291.96
Fruits	1 517	7.43	6.06	0.29	67.40
Vegetables	1 983	12.15	9.79	0.11	81.09

Notes: Data shown are price ratios (price per kilocalorie of edible matter for each food item in each country/year divided by the average price per kilocalorie of basic starchy staples in that country/year) from the World Bank's International Comparison Program (ICP) in 2021. Basic starchy staples are those classified as NOVA 1 (unprocessed or minimally processed foods).

Source: Authors' own elaboration.

Table A5. Number of food items in the International Comparison Program by food group and NOVA category, 2021

	All foods		Unprocessed foods		Processed foods		Ultra-processed foods	
	N	%	N	%	N	%	N	%
Food group								
Oils and fats	1 434	6.11	0	0.00	0	0.00	175	3.34
Starchy staples	4 932	21.01	2 946	23.10	82	2.33	1 904	36.37
Legumes, nuts and seeds	1 120	4.77	574	4.50	464	13.19	0	0.00
Sweets	3 439	14.65	0	0.00	669	19.01	2 184	41.72
Animal-source foods	7 996	34.07	5 276	41.37	1 868	53.08	830	15.85
Fruits	2 257	9.62	1 980	15.52	135	3.84	142	2.71
Vegetables	2 294	9.77	1 978	15.51	301	8.55	0	0.00

Source: Authors' own elaboration.

Table A6. Unique food items in national datasets per metric and category, by country

Category	Mexico	Nigeria	Pakistan
Food Compass Score			
FCS ≤ 20	5 416	32	7
20 < FCS ≤ 40	4 491	15	7
40 < FCS ≤ 60	3 174	27	5
60 < FCS ≤ 80	2 308	32	14
80 < FCS ≤ 100	2 949	45	31
NOVA			
1	4 687	72	42
2	2 040	19	17
3	4 409	27	3
4	7 532	33	4

Source: Authors' own elaboration.

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